WEEK 1

PROGRAM 1

```
#include<iostream>
using namespace std;
int main(){
   cout<<"Zahurunnisa Begum"<<endl;
   cout<<"CSE "<<endl;
   cout<<"25A31A05EI"<<endl;
   cout<<"SEC-D"<<endl;
   return 0;
}
```

PROGRAM 2

```
#include <iostream>
using namespace std;
int main() {
   double num1, num2, num3;
   cout << "Enter three numbers: ";
   :cin >> num1 >> num2 >> num3;
   if (num1 >= num2 && num1 >= num3) {
      cout << "The biggest number is: " << num1 << std::endl;
```

```
    }
    else if (num2 >= num1 && num2 >= num3) {
        cout << "The biggest number is: " << num2 << std::endl;
    }
    else {
        cout << "The biggest number is: " << num3 << std::endl;
    }
    return 0;
}
```

PROGRAM 3

```
#include <iostream>
using namespace std;
int main() {
    char op;
    double num1, num2;
    cout << "Enter an operator (+, -, *, /): ";
    cin >> op;
    cout << "Enter two operands: ";
    cin >> num1 >> num2;
    switch (op) {
```

```
    case '+':
        cout << num1 << " + " << num2 << " = " << num1 + num2;
        break;
    case '-':
        cout << num1 << " - " << num2 << " = " << num1 - num2;
        break;
    case '*':
        cout << num1 << " * " << num2 << " = " << num1 * num2;
        break;
    case '/':
        if (num2 != 0) {
            cout << num1 << " / " << num2 << " = " << num1 / num2;
        } else {
            cout << "Error! Division by zero is not allowed.";
        }
          break;
    default:
        cout << "Error! The operator is not correct.";
        break;
}
cout << endl;
```

```
    return 0;
}
```

PROGRAM 4

```
#include <iostream>
using namespace std;
int main()
{
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << "Prime numbers up to " << n << " are: ";
    for (int i = 2; i <= n; i++)
    {
        bool isPrime = true;
        for (int j = 2; j <= i / 2; j++)
        {
            if (i % j == 0)
            {
                isPrime = false;
                break;
            }
```

```
        }
        if (isPrime)
        {
            cout << i << " ";
        }
    }
return 0;
}
```

PROGRAM 5

```
#include <iostream>
using namespace std;
int main() {
    int n;
    long long factorial = 1;
    std::cout << "Enter a positive integer: ";
    std::cin >> n;
    if (n < 0) {
       cout << "Factorial of a negative number doesn't exist." <<
std::endl;
    } else {
        for (int i = 1; i <= n; ++i) {
            factorial *= i;
```

```
        }
        cout << "Factorial of " << n << " = " << factorial << std::endl;
    }

    return 0;
}
```

PROGRAM 6

```
#include <iostream>
using namespace std;
long long factorial(int n) {

    if (n == 0 || n == 1) {
        return 1;
    }

    else {
        return n * factorial(n - 1);
    }
}

int main() {
```

```
    int number;
    cout << "Enter a non-negative integer: ";
    cin >> number;
    if (number < 0) {
        cout << "Factorial is not defined for negative numbers." <<
std::endl;
    } else {


        long long result = factorial(number);
        cout << "Factorial of " << number << " is: " << result << std::endl;
    }


    return 0;
}
```